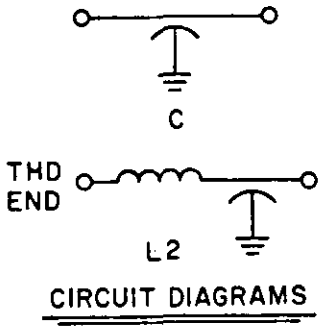
MILITARY SPECIFICATION SHEET

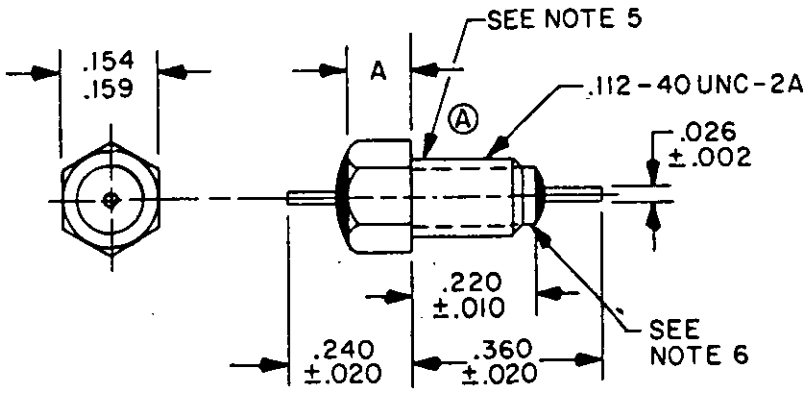
FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, NONHERMETICALLY SEALED, STYLE FS60

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-28861.

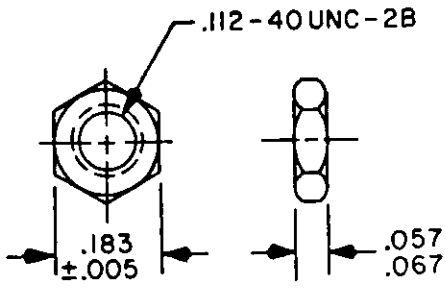


CIRCUIT DIAGRAMS

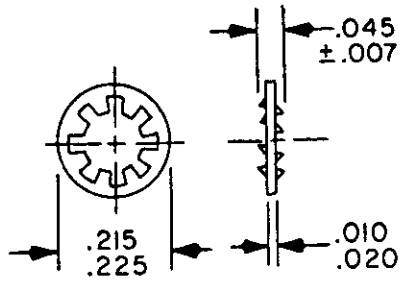


Ⓐ

Dash no.	A dimension
003	.145 ±.030
001, 002 004 thru 007	.120 ±.005



HEX NUT



LOCKWASHER

FIGURE 1. Case and hardware dimensions and circuit diagrams.

Ⓐ denotes changes

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Inches	mm	Inches	mm	Inches	mm
.002	0.05	.045	1.14	.183	4.65
.005	0.13	.057	1.45	.215	5.46
.007	0.18	.067	1.70	.220	5.59
.010	0.25	.112	2.84	.225	5.72
.020	0.51	.120	3.05	.240	6.10
.026	0.66	.154	3.91	.360	9.14
.030	0.76	.159	4.04		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Circuit diagram is for information only.
4. All filters shall be supplied with mounting hardward (hex nut and lockwasher). Mounting hardware shall be furnished with the same finish as the filter case.
Ⓐ5. One and one-half imperfect threads allowed .030 inch maximum.
6. One imperfect thread allowed .030 inch maximum.
7. Recommended mountin torque: 32 oz-in. ±4 oz-in.
8. Potting shall not extend beyond .030 inch from the filter body.

FIGURE 1. Case and hardware dimensions and circuit diagrams - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1 gram maximum.

Ⓐ Case: Steel, UNS 10100 or 10180 per ASTM A108.

Case finish: Class B filters, T only (tin-plated or tin-lead plated).
Class S filters, T or G only (tin-plated, tin-lead plated or gold-plated).
In accordance MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 5 amperes, maximum.

Capacitance to ground: See table I for capacitance value.

Ⓐ Dissipation factor: 3 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1000 megohm-microfarad or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarad or 10,000 megohms minimum, whichever is less.

Insertion loss: At +25°C: In accordance with table I.
At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.1 volt, maximum.

DC resistance: 0.02 ohm, maximum.

Seal: Not applicable.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Product assurance level: See table I.

Part number: The part number shall be as follows:

M28861/06- XXX X X

- XXX: Dash no. per figure 1 and table I
- X: Case finish
- X: Product assurance level per MIL-F-28861

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Ⓐ TABLE I. Electrical characteristics.

Dash number	Circuit	Product assurance level		Rated voltage dc volts	Capacitance (µF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
		Class				1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
		B	S								
001	C	X	X	100	.027	10	30	39	45	65	70
002	L2	X	X	100	.027	10	30	50	54	70	70
003	L2	X		100	.045	14	37	52	58	70	70
004	C	X	X	200	1,000 pF	-	4	20	25	50	55
005	C	X	X	200	5,000 pF	-	15	34	41	50	55
006	C	X		200	.01	4	21	35	42	65	70
007	L2	X		200	.01	4	21	35	38	65	70

1/ Insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz. Insertion loss measurements above this frequency range shall be made under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

Marking: Filters shall be marked, as a minimum, with an abbreviated military part number as shown on figure 2. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.

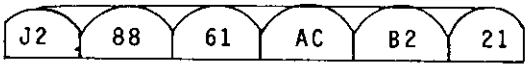


FIGURE 2. Example of marking for the abbreviated military part number on the hex flats - expanded view.

J - Abbreviation for JAN
28861 - Abbreviation for M28861
A - Indicates specification sheet 06
C - Code indicating dash no. as follows:

Code	Dash no.	Code	Dash no.
A	001	E	005
B	002	F	006
C	003	G	007
D	004		

B - Code indicating product assurance level (for example, B indicates class B).
221 - Date code. First digit indicates the year (for example, 2 indicates 1982). The last two digits indicate the week.

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Cataloging information: Dash numbers 001, 004 through 006 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 002, 003, and 007 shall be cataloged under FSC 5915 as radio frequency interference filters.

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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